MERGING SPRINGTIME SPIRALS by Steve Gottlieb

Let's Get TOGETHER

Get your big scope out and feast your eyes on some remarkable distorted galaxies.

Galactic mergers play a key role in the formation and evolution of galaxies, with the most spectacular examples involving massive, gas-rich spirals. When galaxies resembling the Milky Way collide, gravitational tides fracture the delicate pinwheels, producing dramatic bridges, plumes, and tails that stretch for more than 100 million light-years. The real action, though, occurs in the centers of these galaxies, which are deeply buried inside a shroud of obscuring dust. Despite this barrier, astronomers have succeeded in modeling the evolution of spiral mergers using computer simulations and multiwavelength studies.

During a smash-up, galactic bars funnel a massive inflow of molecular gas and dust to the region of the nucleus. The gas can ignite a nuclear starburst, or it may trigger an *active galactic nucleus* (AGN) by feeding a gas-guzzling *supermassive black hole* (SMBH). The surrounding dust absorbs most of the optical and ultraviolet radiation produced by newborn massive stars or emitted by the SMBH's accretion disk. This energy is re-emitted as heat, boosting the galaxy's infrared luminosity. In a *luminous infrared galaxy* (LIRG), the output surpasses 100 billion times the Sun's luminosity.

As a merger advances, the dust-enshrouded nucleus launches a fast outflow of heated gas. In a feedback process, the energy can both spark and stifle star formation, and regulate the feeding of the SMBH. These episodic processes occur in the nuclei of both merging galaxies. Eventually, the coalescing merger is depleted of dust and gas — some is blown out into intergalactic space — and star production is

▲ **ARP 302** These two merging galaxies in Boötes, both gas-rich spirals, are at the very early stages of their interaction. The Hubble Space Telescope image beautifully reveals that the galaxy at lower left (VV 340b) is face-on while the one at upper right (VV 340a) is edge-on.

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shut down. The metamorphosis results in a massive, gas-poor elliptical galaxy.

Astronomers identify five stages in a major merger, involving two or three close encounters over a billion-year time frame. Several details remain poorly understood, but future research with the James Webb Space Telescope, which operates in the dust-busting infrared, should help resolve several open questions. The stages are:

M1: pre-merger pair on their first approach with no prominent tidal features.

M2: early-stage interacting pair with obvious tidal tails and a bridge.

M3: mid-stage with multiple nuclei in a disturbed overlapping disk, with visible tidal tails.

M4: late-stage with a single nucleus and some remaining tidal features.

M5: coalescent remnant with a diffuse envelope and perhaps shells, but no tidal trauma.

Let's tour seven mergers well placed in springtime that highlight the stages in order from M1 through M4, with views through my 18-inch and a couple of mega-size telescopes.

The Sole M1

Arp 302 (VV 340) is an early M1 interaction involving two gas-rich spirals of equal magnitude, with their cores 38″ apart. The northern galaxy's infrared emission surpasses 500 billion times that of our Sun's, while massive star clusters line the spiral arms of the southern galaxy. Arp 302 is located 3.6° southeast of 2.4-magnitude Epsilon (ε) Boötis, also known as Izar, a gorgeous pair separated by 2.9″ with a 2.6-magnitude yellow-orange primary and 4.8-magnitude secondary. The 6.9-magnitude star HD 132304 is 10′ northeast of the galaxies and provides a handy signpost.

I found a single extended glow at 175× with my 18-inch reflector, but boosting the magnification to 285× split the duo. The northern edge-on appears as a fairly thin 30″ slash. The face-on spiral hovers just to its south as a hazy circular glow, about 20″ in diameter. A 15th-magnitude star at its northern edge separates the two galaxies.

USEFUL SOURCES

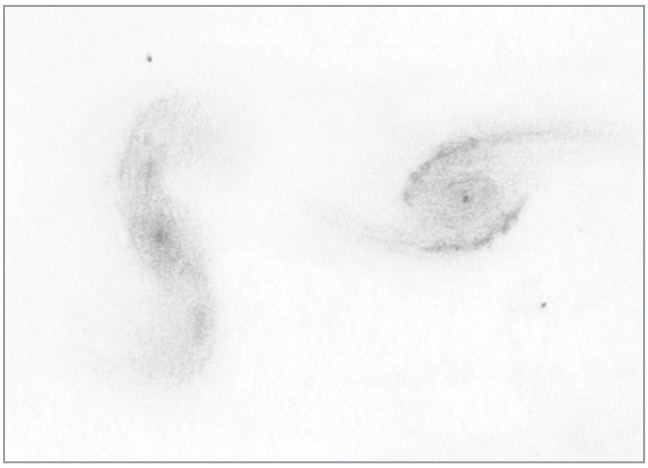
Halton Arp's *Atlas of Peculiar Galaxies* (1966) and Boris Vorontsov-Velyaminov's *The Atlas and Catalogue of Interacting Galaxies* (1959) contain many examples of photogenic M2 and M3 mergers such as The Antennae (NGC 4038-39) and The Mice (NGC 4676). The names of objects in Vorontsov-Velyaminov's catalog begin with VV.

A Pair of M2s

NGC 5257 and NGC 5258 form **Arp 240**, an elegant M2 pair of LIRGs connected by a filament of dust and gas. Simulations show that their closest approach occurred about 250 million years ago. To find this pair, point to 3.4-magnitude Zeta (ζ) Virginis and slide 2° to the northeast.

Viewed at 175× in my 18-inch, the two spirals are similar in size and brightness, with weakly condensed cores. NGC 5258 elongates 5:2 southwest to northeast and runs 1′ in length. Using 285×, a 15.5-magnitude star popped into view at the northern edge. NGC 5257 spans 0.9′ × 0.6′ and slopes perpendicular to its companion. A brighter, 14th-magnitude star sits off its western flank.

When I observe with Jimi Lowrey and use his superb 48-inch telescope, interacting pairs are always high on our observing list — this stunning duo illustrates why. At 610×, NGC 5258 flaunts its two spiral arms. The main body stretches 3:1 and houses a small bright core. I noticed an H II region midway between the core and the northern tip along the major axis. A strong spiral arm emerges at the southwestern end and hooks sharply to the east. A much fainter arm



▲ **ARP 240** This glorious duo in Virgo comprises NGC 5257 (right) and NGC 5258 (left), spirals that are very similar both in mass and size. A dim bridge of gas, dust, and stars connects the pair. Howard Banich sketched the view at the eyepiece of Jimi Lowrey's 48-inch at magnification 690×.

GALAXY ICONS: SUPRIYA07 / SHUTTERSTOCK.COM

▲ **ARP 238** Head over to Ursa Major to find this pair, heavily distorted by tidal forces. Both galaxies exhibit dust lanes at their centers.

curls west from the northeast side, passing the 15.5-magnitude star, and points towards its companion.

Switching my attention to NGC 5257, I noted thin, high-surface-brightness arcs bordering its southwestern and northeastern flanks. Both arcs are convex and form a striking pair of parentheses enclosing the core. With closer scrutiny, thin dust lanes line the insides where the surface brightness drops dramatically. The northern parenthesis appeared to lengthen into a spiral arm to the west-northwest, stopping near a 15th-magnitude star. The southern parenthesis blends into a narrow, dim bridge towards NGC 5258 but falls just short of the companion.

Arp 238 (VV 250) is a tight M2 pair locked in a gravitational embrace. It's a great example of the havoc wreaked by a strong tidal grip. A gas and dust bridge entangles the duo, and each galaxy displays wildly contorted tails. In addition, deep images show fountains of freshly minted hot blue stars that appear ripped out of the plane of the southeastern spiral.

This diminutive twosome lies 6° southwest of 3.7-magnitude Thuban, or Alpha (α) Draconis, in Ursa Major and 40′ east of 6.5-magnitude HD 114504. In my 18-inch, just a tiny gap (35″ between their centers) separates the pair. The slightly brighter southeastern galaxy spans 20″, while its intertwined partner is a dim 15″ smudge. Some 30″ northeast is a duo of 14th-magnitude stars that mimic the orientation and separation of the galaxies.

The 48-inch showed the western tidal tail as a low-surface-brightness extension with a sharp hook to the north. The eastern member has a bright core that intensified toward the center; I spotted its debris tail as a feeble arc bending south. Completing the scene, a luminous bridge of misty haze links the pair.

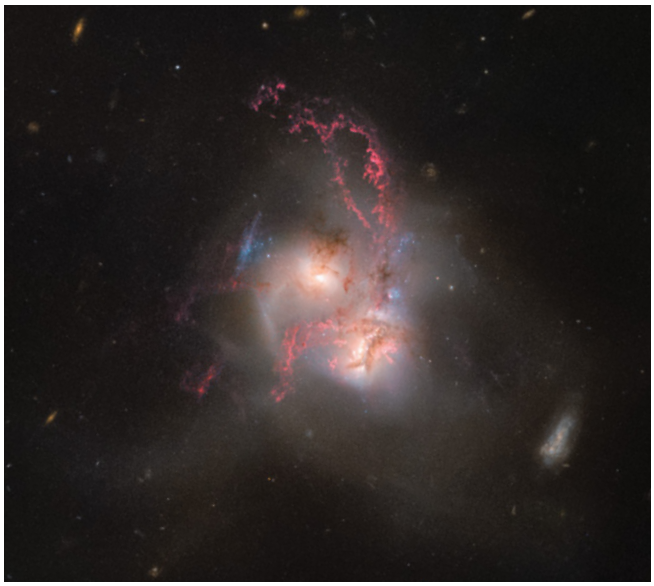
A Trio of M3s

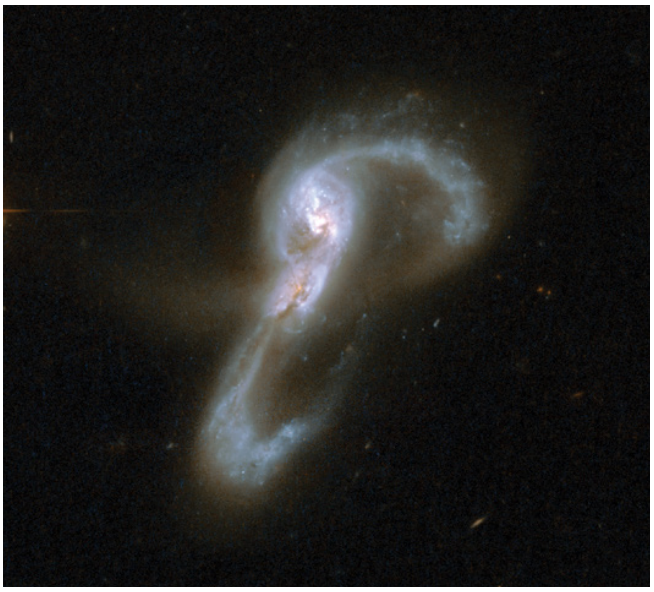
Arp 299 (NGC 3690), which lies in the bowl of the Big Dipper, is a deformed M3 crash. It ranks as the brightest galaxy in the infrared within 150 million light-years. Enveloped in its single halo are two nuclei separated by 20″, and a frenzy of star formation just north of the western nucleus. Although most of the infrared emission originates from the dust-enshrouded eastern nucleus, both may contain a composite AGN/nuclear starburst. Arp 299 is producing supernovae at a furious rate — at least seven have exploded since 1992.

Springtime Mergers

Object	Other designation	Type	Dist. (M l-y)	Mag(v)	Sep. Nuclei	Size	RA	Dec.
Arp 302	VV 340a; UGC 9618N	M1	460	~14.5	38″	0.8′ × 0.2′	14^h 57.0^m	+24° 37′
	VV 340b; UGC 9618S			~14.5		0.6′ × 0.6′	14^h 57.0^m	+24° 36′
Arp 240	NGC 5257	M2	315	12.4	1.4′	1.6′ × 0.8′	13^h 39.9^m	+00° 50′
	NGC 5258			12.3		1.4′ × 0.9′	13^h 40.0^m	+00° 50′
Arp 238	VV 250	M2	435	~14.0	36″	1.7′ × 0.7′	13^h 15.5^m	+62° 08′
Arp 299	NGC 3690W	M3	145	11.2	20″	1.6′ × 1.4′	11^h 28.5^m	+58° 34′
	NGC 3690E			10.9		2.0′ × 1.4′	11^h 28.6^m	+58° 34′
NGC 5256	Mrk 266	M3	390	13.2	10″	1.2′ × 1.1′	13^h 38.3^m	+48° 17′
VV 705	Mrk 848	M3	560	~15	6″	0.8′ × 0.4′	15^h 18.1^m	+42° 45′
Mrk 273	VV 851; UGC 8696	M4	520	14.9	1″	1.1′ × 0.3′	13^h 44.7^m	+55° 53′

Angular sizes and separations are from recent catalogs. Visually, an object's size is often smaller than the cataloged value and varies according to the aperture and magnification of the viewing instrument. Right ascension and declination are for equinox 2000.0.





◀▲ **ARP 299** Staying in Ursa Major, we find this duo in which a close encounter some 700 million years ago triggered a burst of star formation. Various sources misidentify Arp 299 as NGC 3690 and IC 694, or apply the latter designation to the eastern component. Instead, IC 694 is a 16th-magnitude elliptical 1′ to the northwest. Observers using the 72-inch Leviathan reflector of William Parsons, the Third Earl of Rosse, first noted this dim speck in 1852.

▲ **NGC 5256** This twosome, also in Ursa Major, dramatically shows the effects of merging: The red streamerlike features are gas and dust spilled from the guts of the galaxies, while the blue patches are regions of recent star formation.

◀ **VV 705** Back in Boötes, we come to this tight pair — the cores of these galaxies are only some 16,000 light-years apart; in fact, the system is likely midway through a merger.

Using my 18-inch at 285×, the bright western half (NGC 3690W) encompasses a conspicuous starlike nucleus, while the fainter eastern member (NGC 3690E) shows little concentration. A common irregular envelope, which spans 1.25′ in diameter, enfolds the pair.

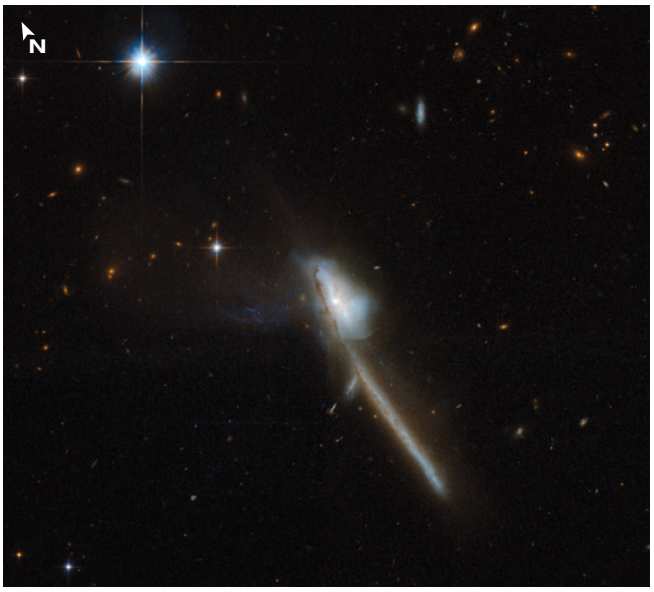
At 610× in Jimi's scope, NGC 3690W was a dazzling, irregular knot of high surface brightness with a brilliant quasi-stellar nucleus. Larger NGC 3690E held a small vivid core. A very low-surface-brightness, asymmetric halo spread northwest, and a detached H II region briefly materialized off its edge. Hovering nearby are two more galaxies. I easily viewed IC 694, a 16th-magnitude companion 1′ northwest, along with Arp 296, an interacting pair 2.5′ northeast that lies far in the background.

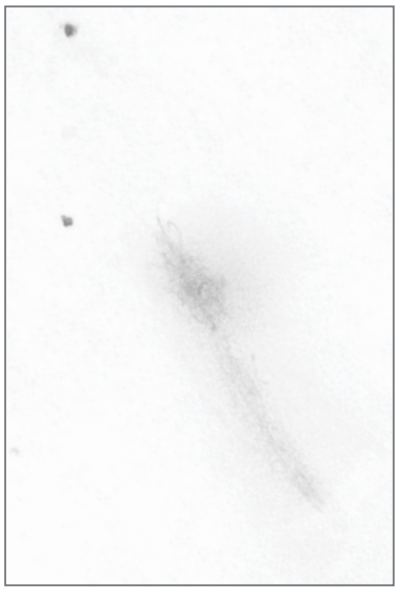
NGC 5256 is a mid-merger (M3) train wreck hosting a dual AGN separated by just 15,000 light-years. The SMBHs each weigh in at a hefty 200 million solar masses. According to a 2012 study at infrared and X-ray wavelengths, a fierce "superwind" is driving an intense outflow of gas and dust, forming ionized ribbons of nebulosity to the north.

To locate NGC 5256, point to 1.9-magnitude Alkaid, Eta (η) Ursae Majoris, the end star in the handle of the Big Dipper. Our target is just 1.8° southwest of the star, midway between Alkaid and the showpiece Whirlpool Galaxy (M51)!

In my 18-inch, I found a moderately bright 40″ × 30″ oval, leaning southwest to northeast. The surface brightness appeared uneven, though I couldn't resolve the dual nuclei, a mere 10″ apart. But jumping to 375× in my 24-inch exposed the quasi-stellar eastern nucleus. Viewed with Jimi's 48-inch, a fainter outer halo encased the luminous central region, and I easily resolved both nuclei.

At a distance of 560 million light-years, **VV 705** is the most remote M3 pair on our tour. So there's no surprise a whisker-thin 6″ separates its nuclei. This dusty LIRG sports a pair of looping tidal arms that make a wild U-turn and even overlap each other. A 2019 spectroscopic study detected





◀ **MRK 273** In May 2019, the author and Banich observed Mrk 273 in a private viewing session with fellow amateur and telescope operator Jim Chandler. This was at the end of a weekend celebration of the 80th anniversary of the 82-inch Otto Struve telescope at McDonald Observatory, which at dedication was the second-largest in the world. For two and a half hours they were treated to views of objects of their choosing. Banich sketched this view of Mrk 273 through the eyepiece of the 82-inch at magnification 617×.

a galactic-wide outflow of gas driven by either AGN or starburst activity.

To find VV 705, look 3.8° northeast of 3.5-magnitude Beta (β) Boötis, the star marking the head of the Herdsman. My 24-inch displays a small oval tilting north-northwest with a tiny but conspicuous nucleus offset to the eastern side. I had no luck seeing two cores, at least not at 282×.

The 48-inch cleanly split the two tight cores at 610×, with the brighter northern one increasing to a stellar peak. I caught the northern tidal tail as a hairline arc bending clockwise towards the west. The larger southern tail brightened and widened before turning a short distance west.

The Forlorn M4

The monsters of the gas-rich mergers are *ultraluminous infrared galaxies* (ULIRGs). Their prodigious quasarlike luminosity exceeds 1 trillion Suns and 100 times the entire Milky Way. ULIRGs form during the final merger stages (M4 and M5) and contain voraciously feeding SMBHs.

Our last stop is **Markarian 273** in Ursa Major, the second-closest ULIRG at 520 million light-years away. (The nearest is Arp 220, or IC 4553, in Serpens; at a distance of 250 million light-years, it emits 99% of its energy in the far-infrared.) As a relatively nearby laboratory, this M4-class powerhouse is a favorite target for researchers, with 854 (and counting) references in the SIMBAD bibliographic database. The power source is a dual AGN separated by less than 2,500 light-years. The southwest nucleus is a *Type II Seyfert galaxy* with narrow emission lines, while the northern nucleus — a scant 1″ away — contains an extreme starburst producing 60 solar masses per year. A slender tidal tail stretches south for 130,000 light-years with fainter diffuse plumes extending a similar distance north and east.

To reach Mrk 273, you can sweep 3° east-northeast of 2.2-magnitude Mizar, Zeta Ursae Majoris, or head the same distance northwest of M101. Unfortunately, our target lies 4′ west of the glare of 6.5-magnitude HD 119992, so park this luminary outside your eyepiece field. My 18-inch shows a pale gray oval running north to south, and my 24-inch provides tantalizing glimpses of the southern tail.

In May 2019, Howard Banich and I had a stunning view of Mrk 273 through the 82-inch Otto Struve telescope at McDonald Observatory. At 617×, the mashed-up body intensified to an elongated core with a vivid nucleus. But the real thrill was its prominent tail, which looked like a jet contrail streaming away from Mrk 273!

Keep in mind that when we observe these spiral mergers, we're getting a sneak preview of our own galaxy's future destiny. The Milky Way is on an inevitable collision course and eventual merger with the Andromeda Galaxy (M31) in several billion years.

EXTREMES

Even more exotic are *hyper-luminous infrared galaxies* (HyLIRGs), which top 10 trillion times (10^{13}) the Sun's luminosity. The current titleholder for the most prodigious output is WISE J224607.55-052634.9, discovered in 2015. It blazed in the very early universe (light-travel time of 12.5 billion years) with an infrared luminosity of 220 trillion Suns.

■ Contributing Editor **STEVE GOTTLIEB** chases colliding LIRGs and other exotic targets from northern California. He can be reached at **astrogottlieb@gmail.com**.

EXTRA MATERIAL Go to **https://is.gd/merging_spirals** for finder charts for the targets discussed here.

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